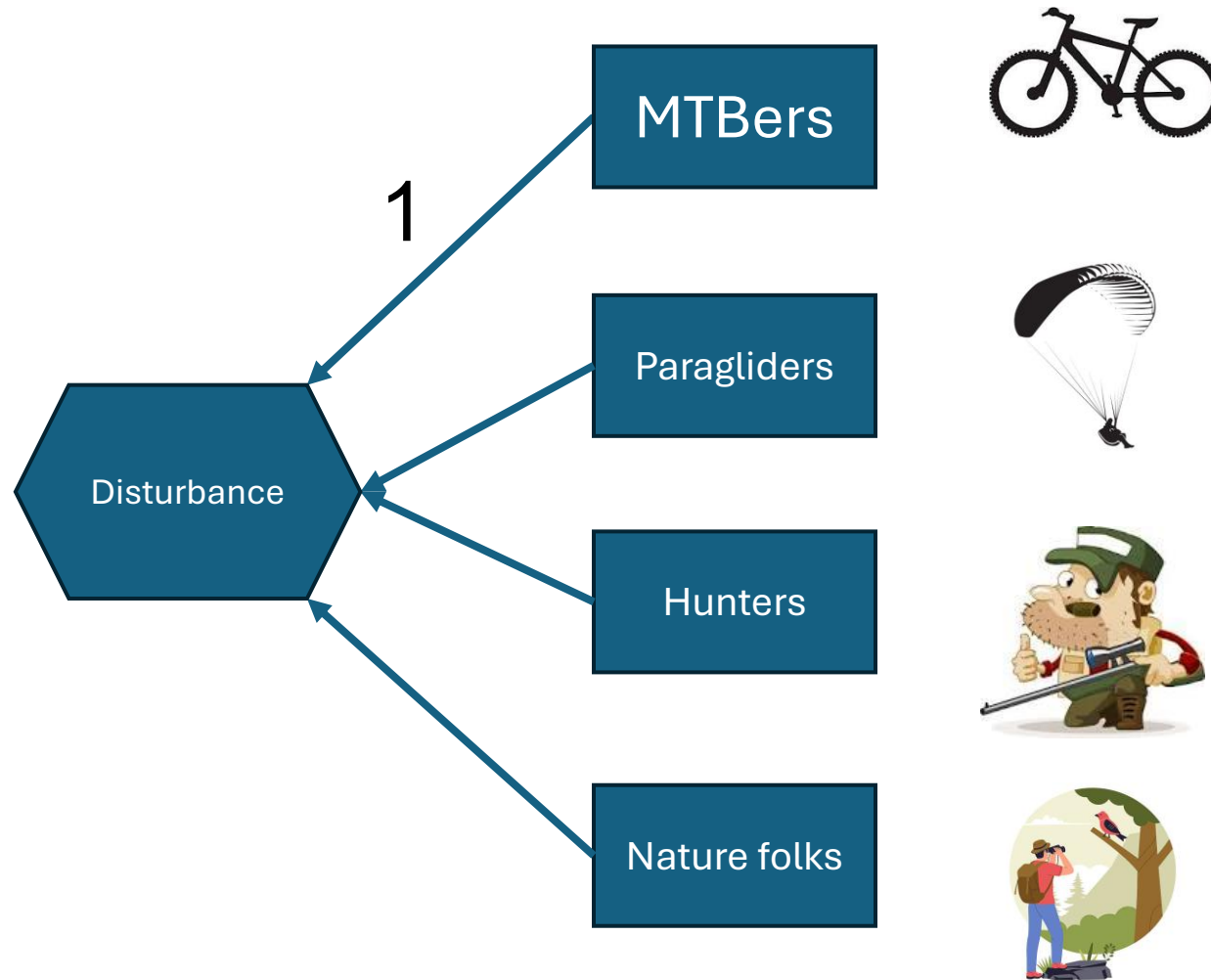


Composite variables



What is a latent (or hidden) variable?



A random variable that is unmeasured but not necessarily unmeasurable.

-P Spirtes (2001)

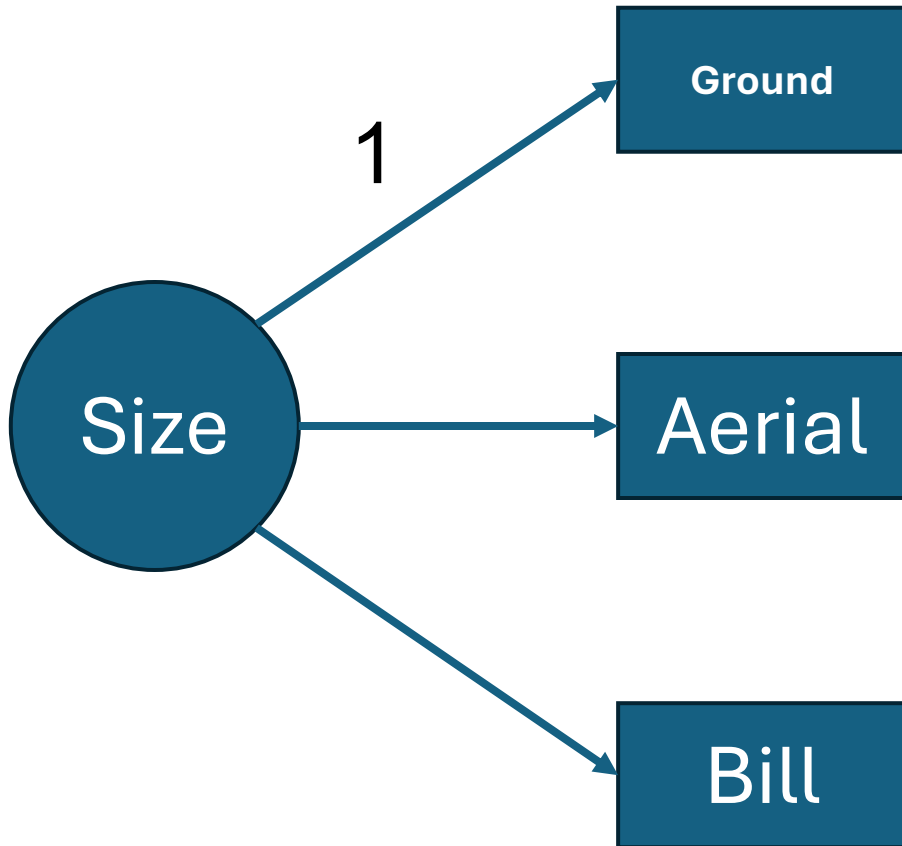
A variable that is not directly observable but is inferred from other variables that can be measured

-Generative AI (yesterday)

Variables that can only be inferred indirectly through a mathematical model from other observable variables

-Wikipedia (also yesterday)

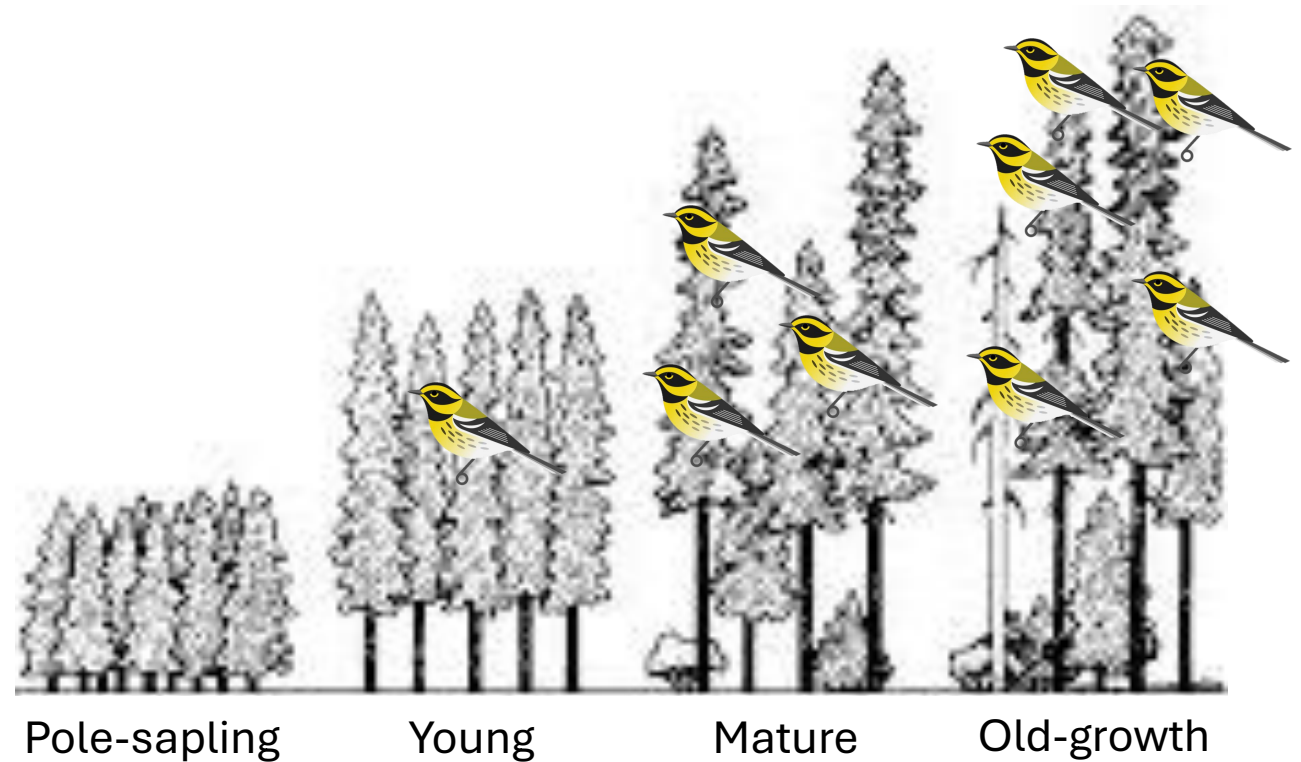
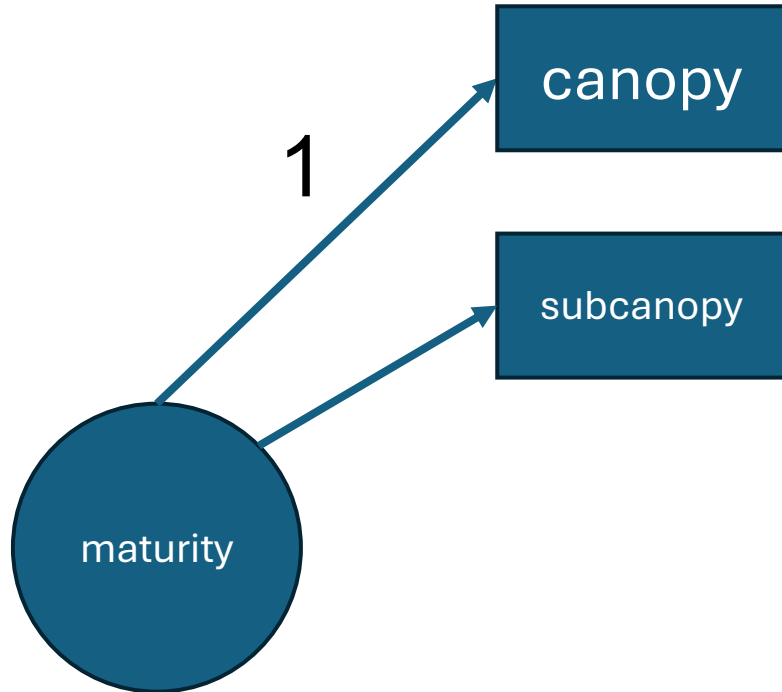
Size



‘Size’ is a human construct



Forest maturity



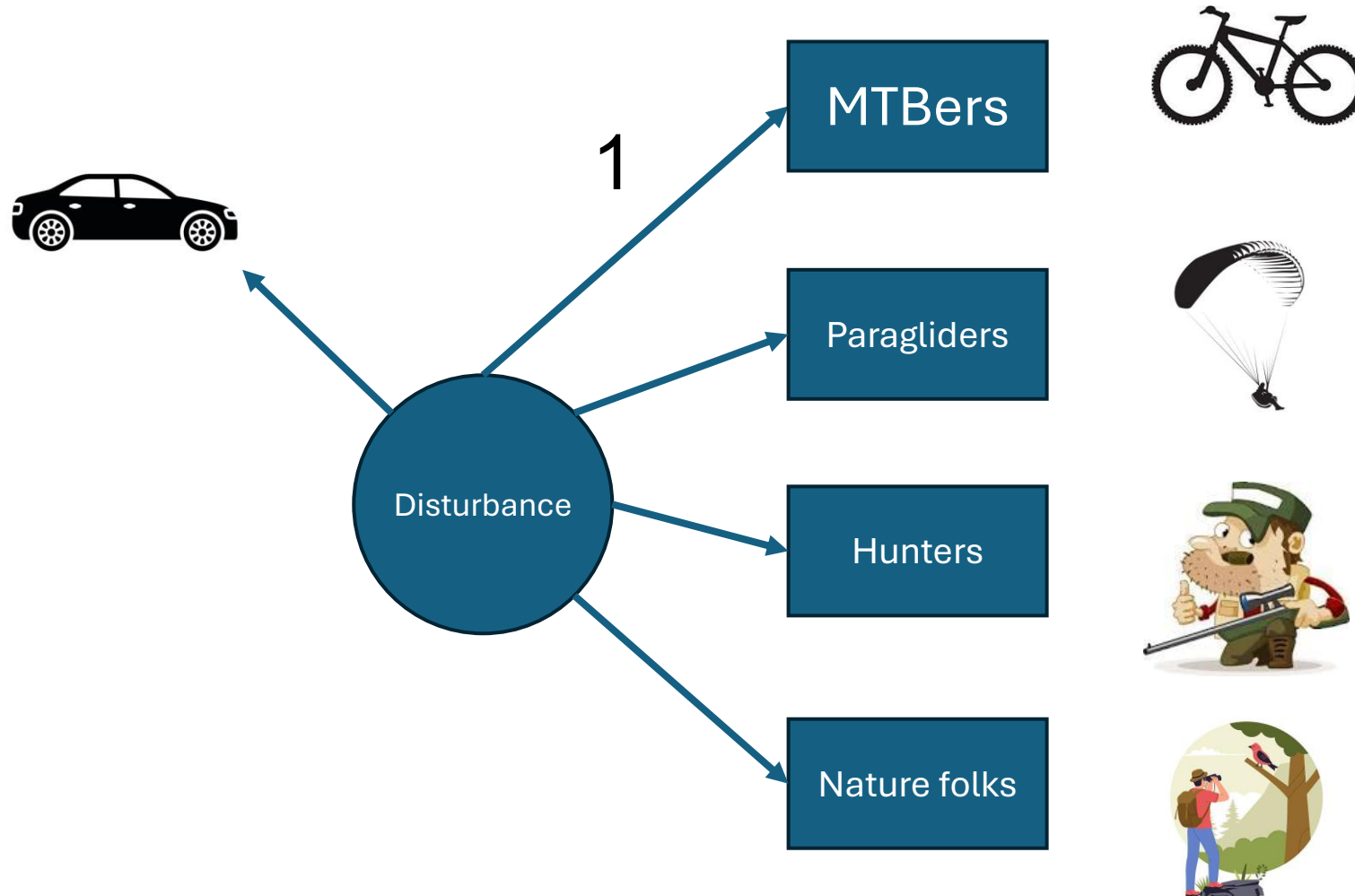
These 'seral stages' are human constructs

What is a latent (or hidden) variable?



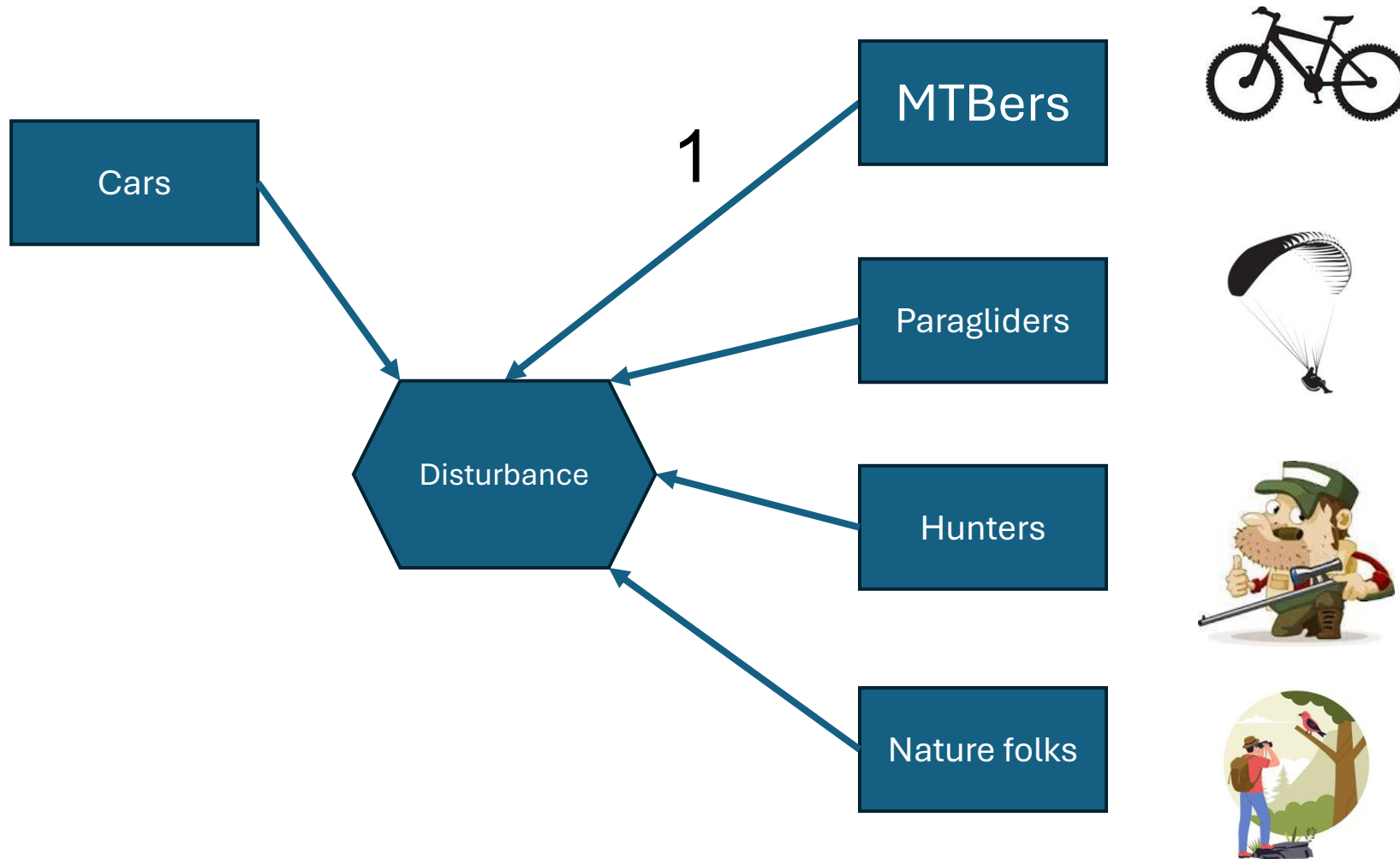
Everything is a latent variable – **some pedant**

Is disturbance a latent variable



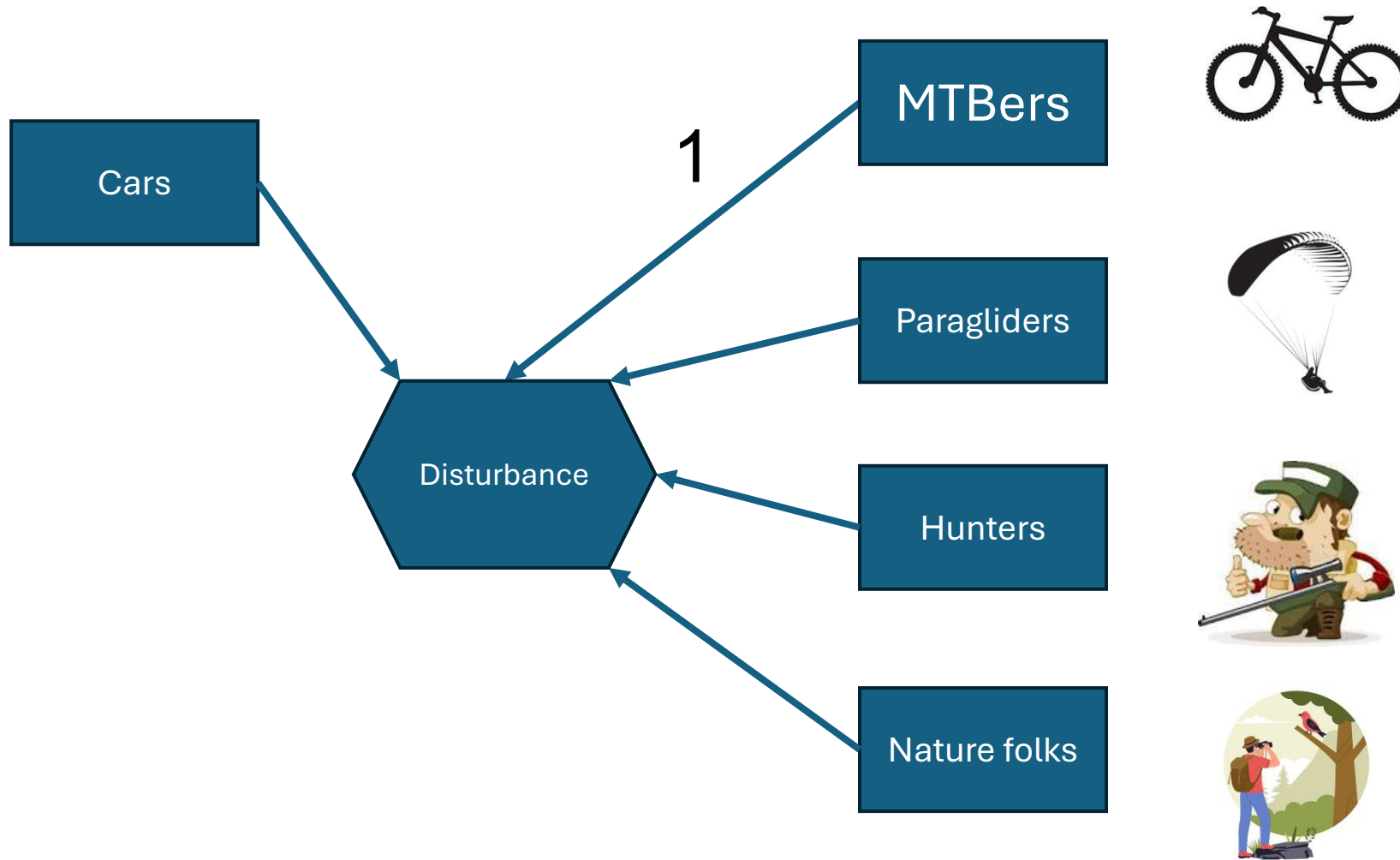
Disturbance is a human construct (+roads), but...

Composite variables: flipping the arrows

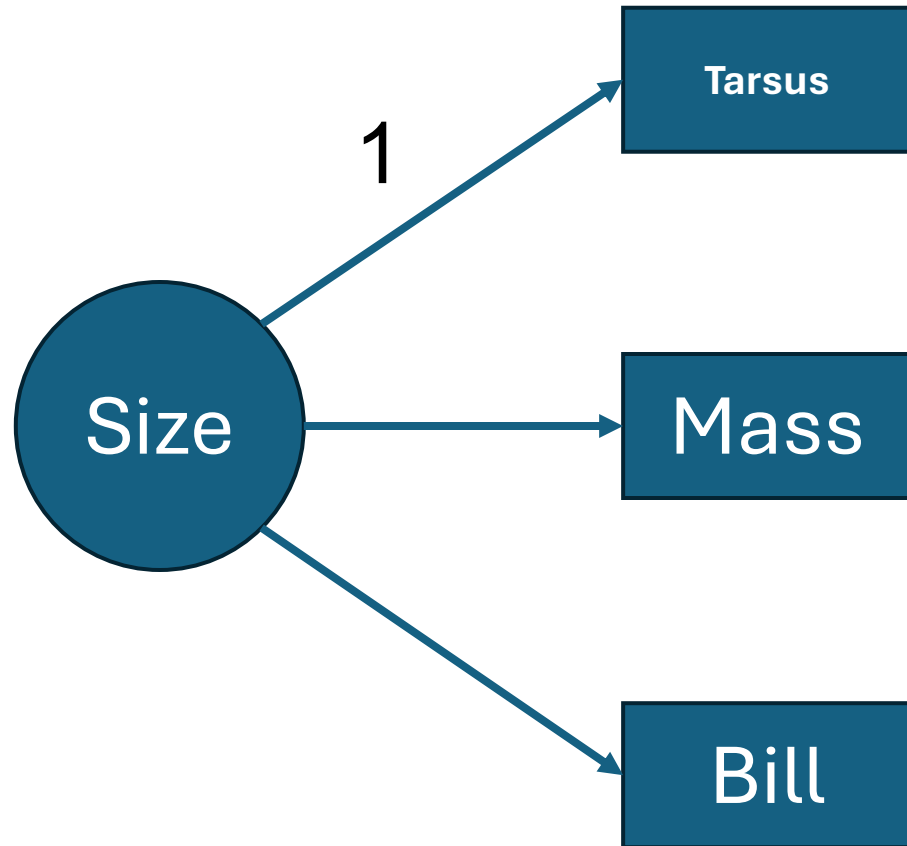


All forms of human disturbance combine to create ‘disturbance’?

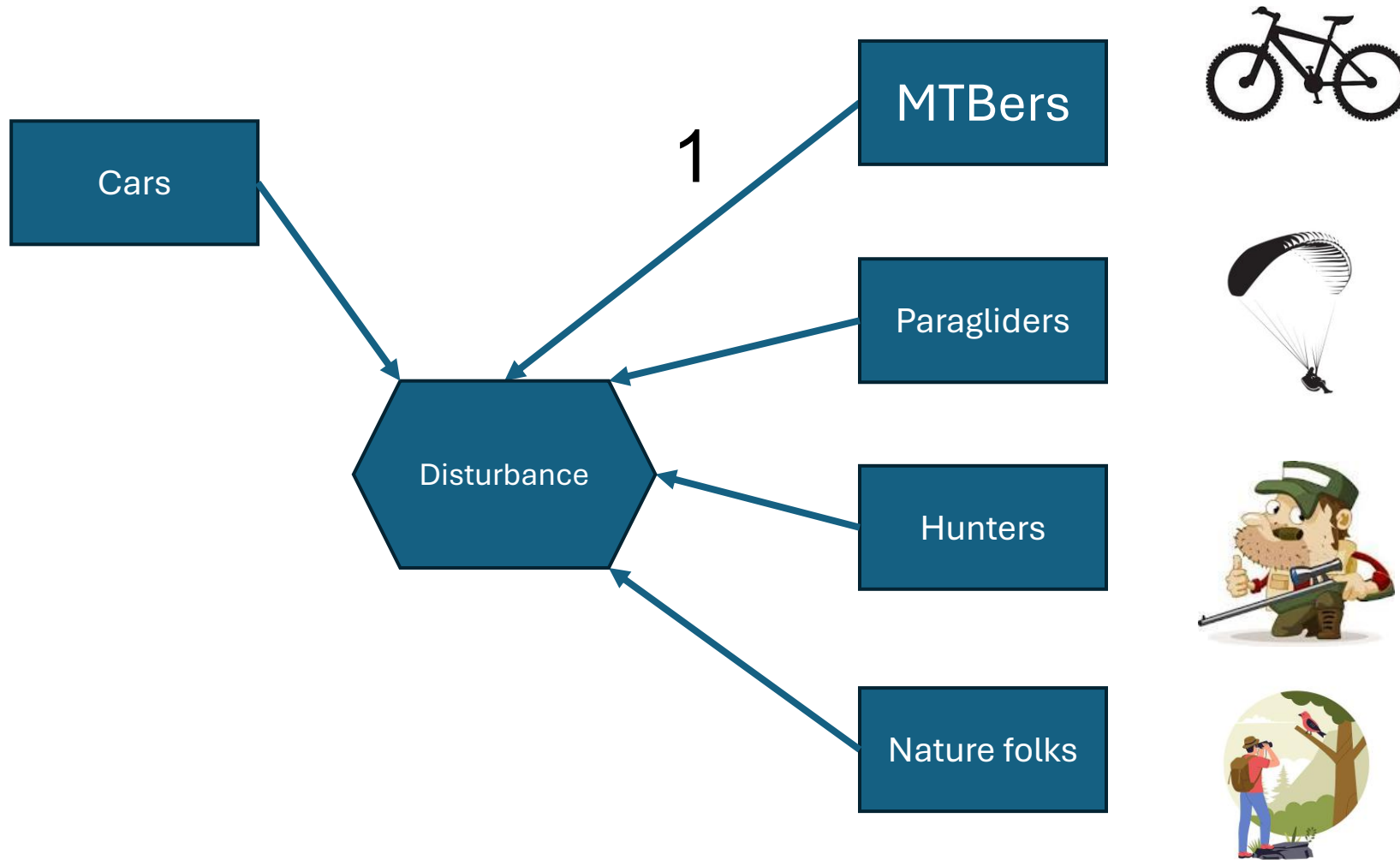
We often use proxies, e.g., distance to road? road density?



These measured variables have to be collinear



Do these measured variables have to be collinear?



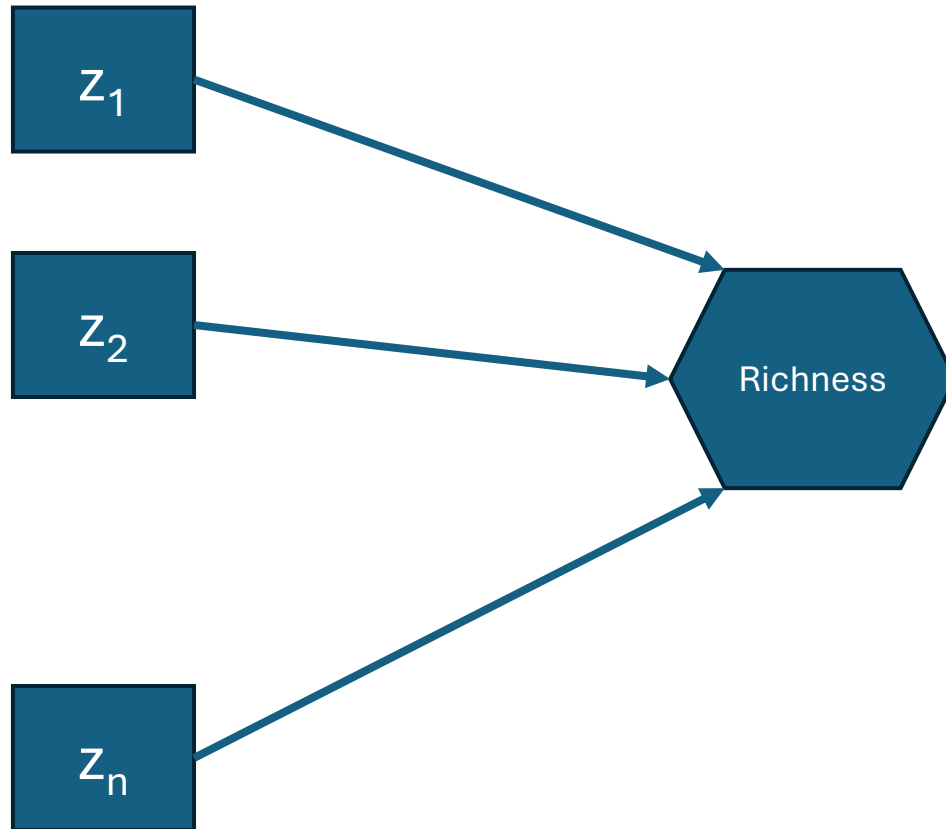
What is a composite variable?



***Latent variables** give rise to measurable manifestations of an unobservable concept, while **composite variables** arise from the total combined influence of measured variables*

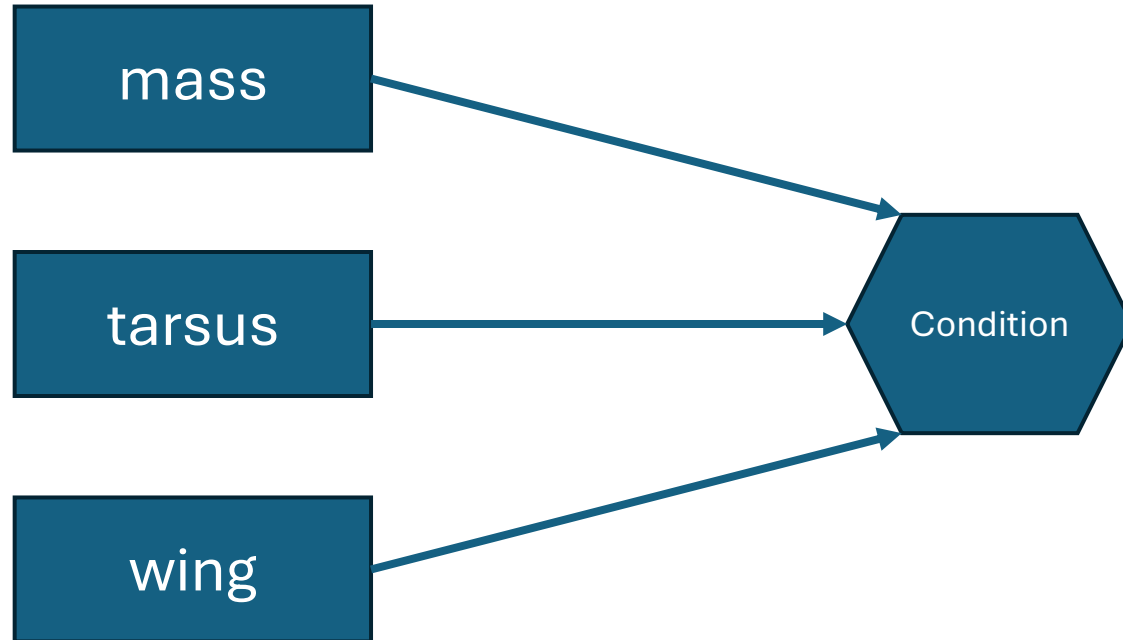
- Jon Lefcheck

E.g., species richness



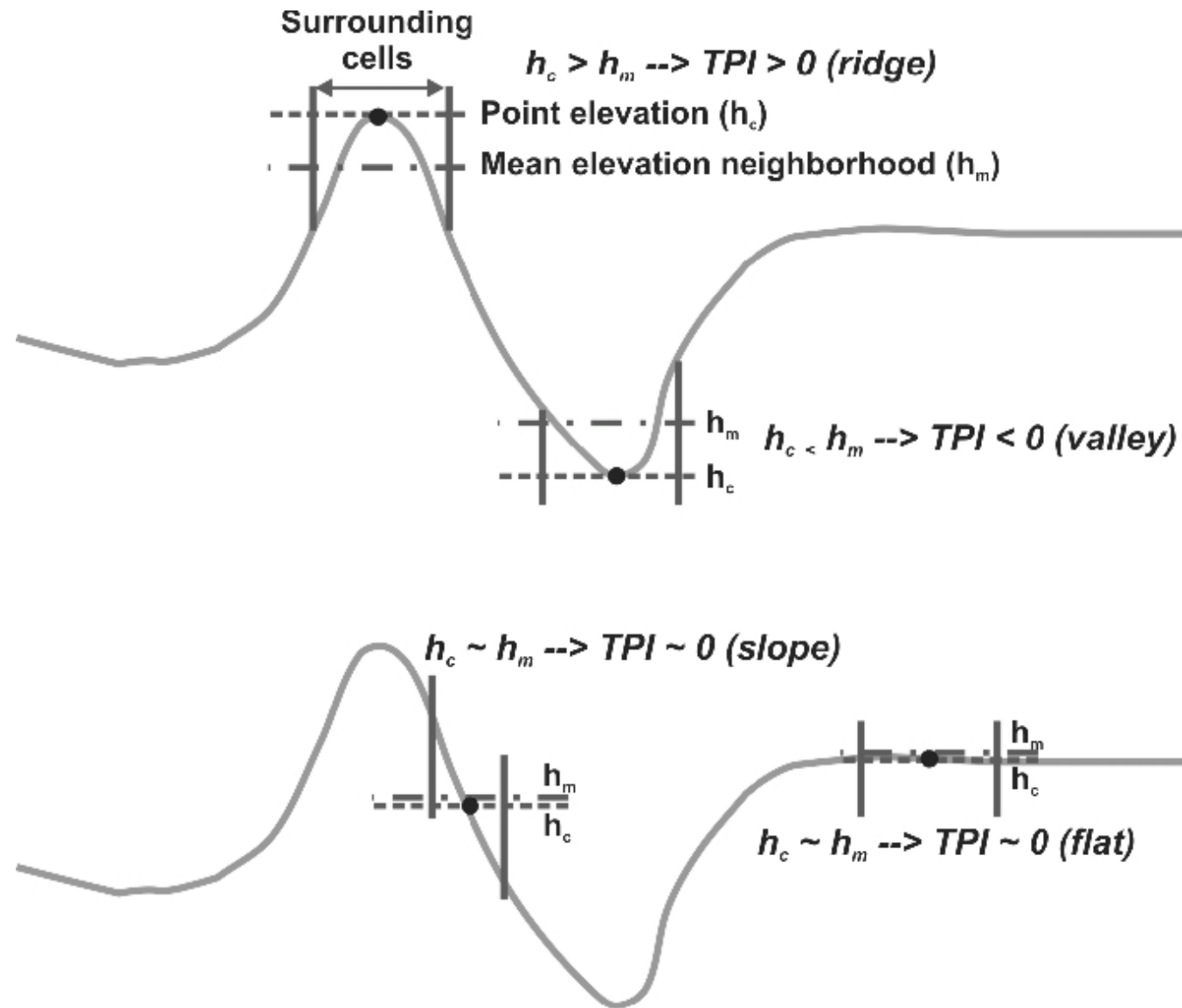
Species richness is the sum of species presence for n species

E.g., body condition



Body condition is a function of weight and structural measurements

E.g., Topographic position index



E.g., Terrain ruggedness index



$$TRI = \sqrt{\sum_{i=1}^8 (x_i - E)^2}$$

where x_i refers to each of the eight neighbors of the center cell E .

Without loss of generality, suppose we are calculating the TRI using only the four nearest neighbors to the center cell, or cells labeled B, D, F, and H. Applying the formula, the TRI would be:

$$TRI = \sqrt{(B - E)^2 + (D - E)^2 + (F - E)^2 + (H - E)^2}$$

Expanding the term inside the square root, we have:

$$B^2 - 2BE + E^2 + D^2 - 2DE + E^2 + F^2 - 2FE + E^2 + H^2 - 2HE + E^2$$

Table 1: 3-by-3 kernel centered on cell E

A	B	C
D	(E)	F
G	H	I

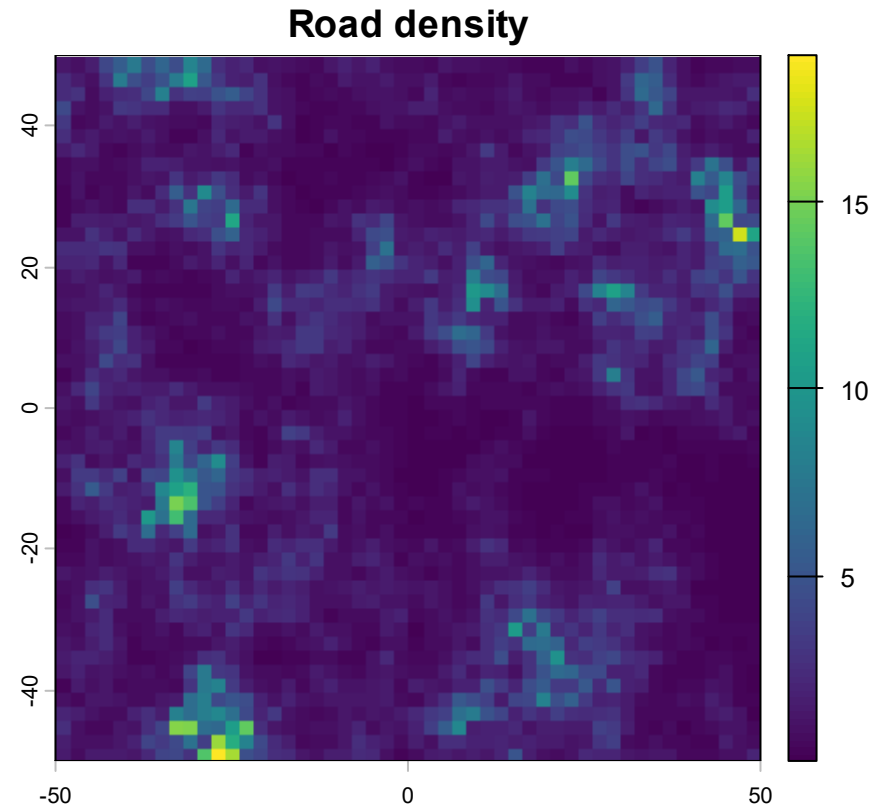
What is a composite variable?



We make composite variables all the time...

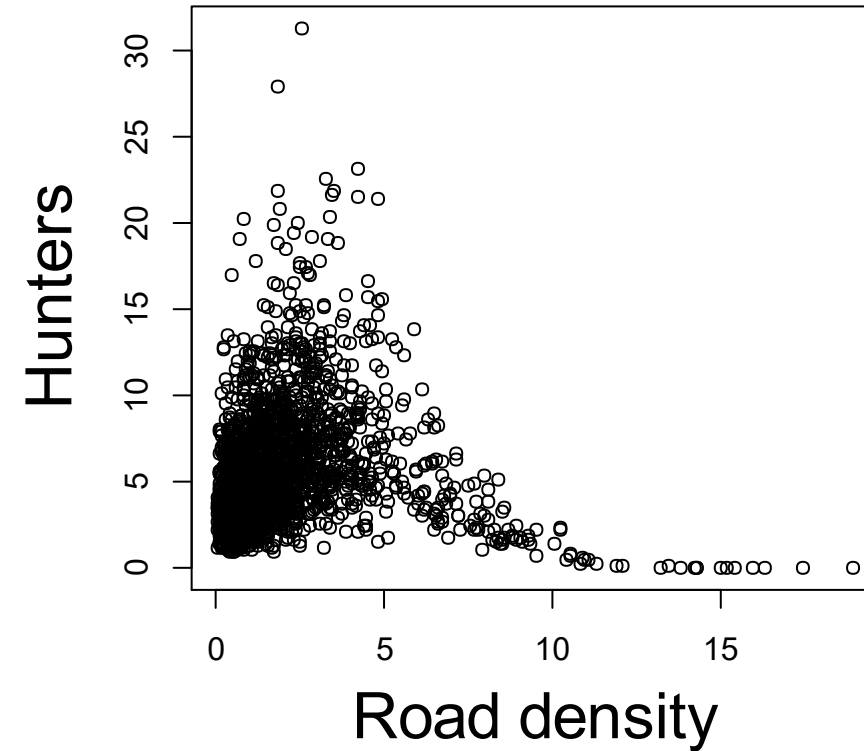
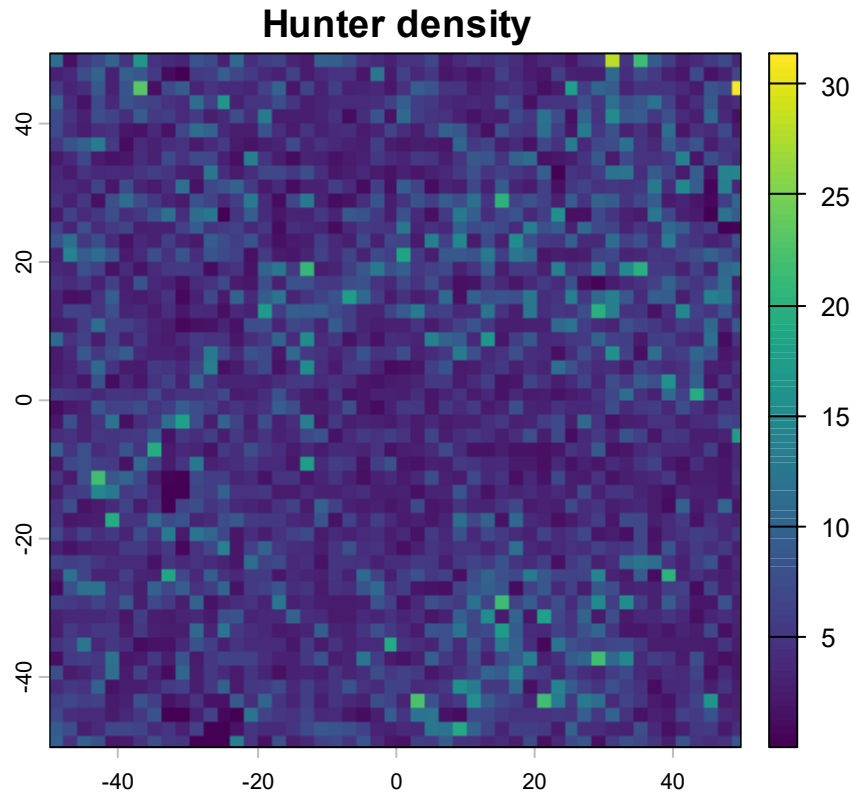
We just do it outside of the model and then input it as a covariate with no measurement error

Let's simulate some data



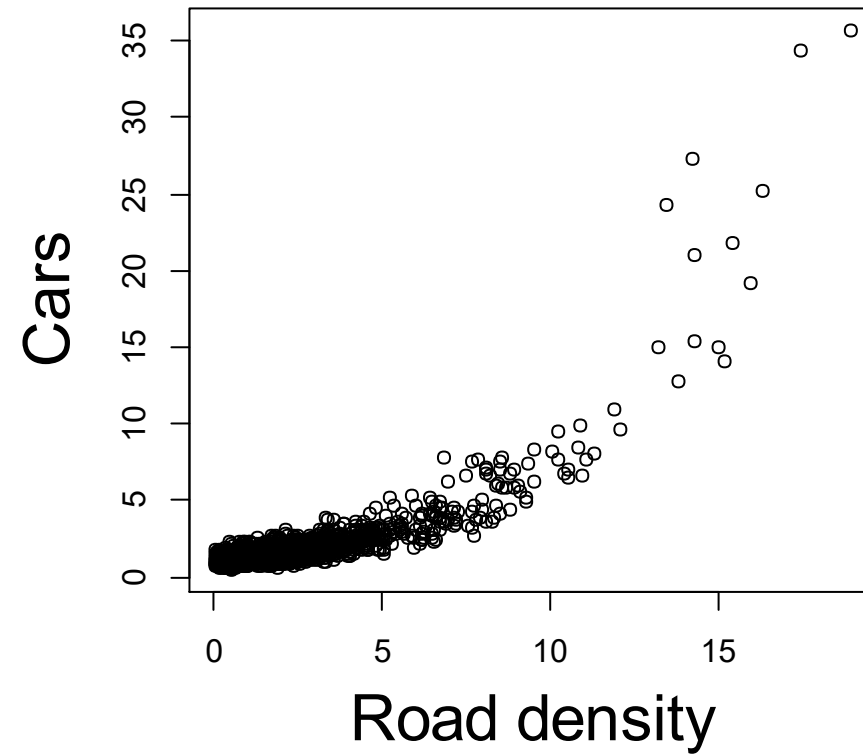
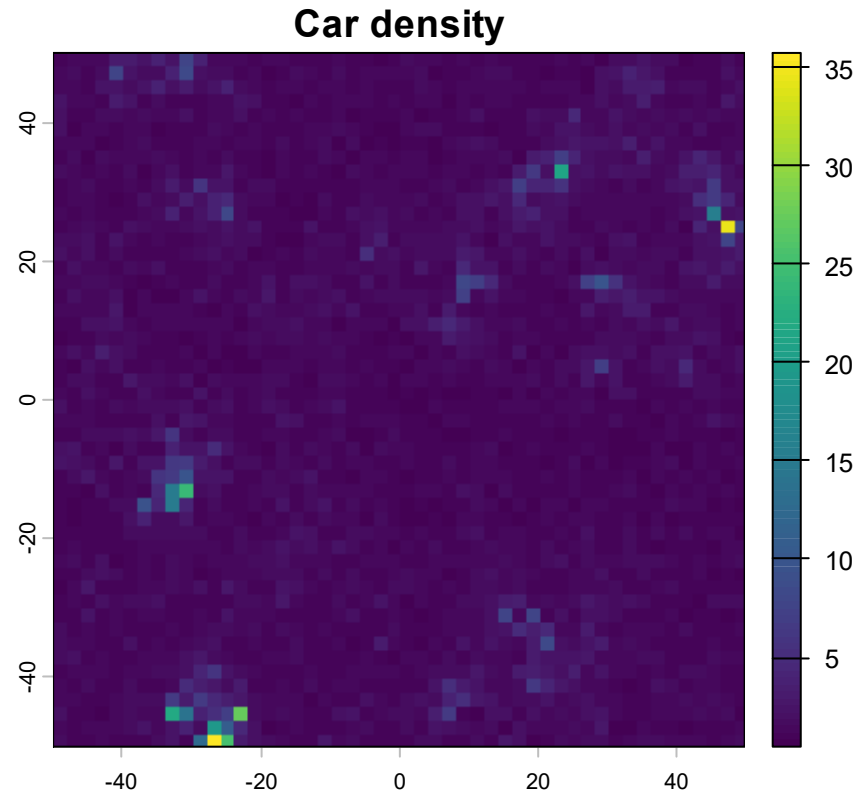
We'll begin by simulating road density as a Gaussian random field

We'll simulate hunter density



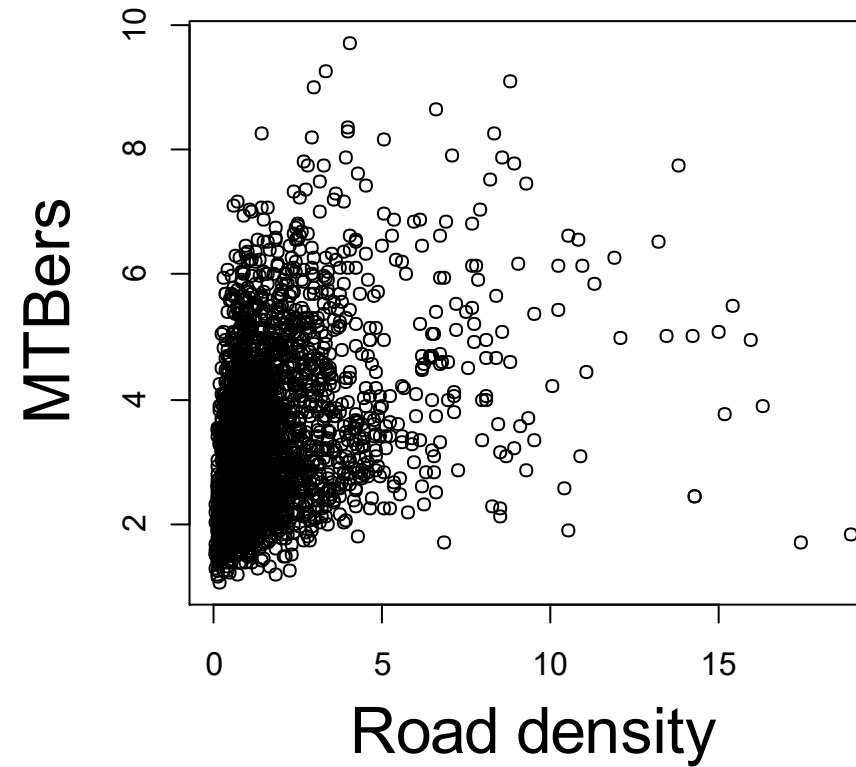
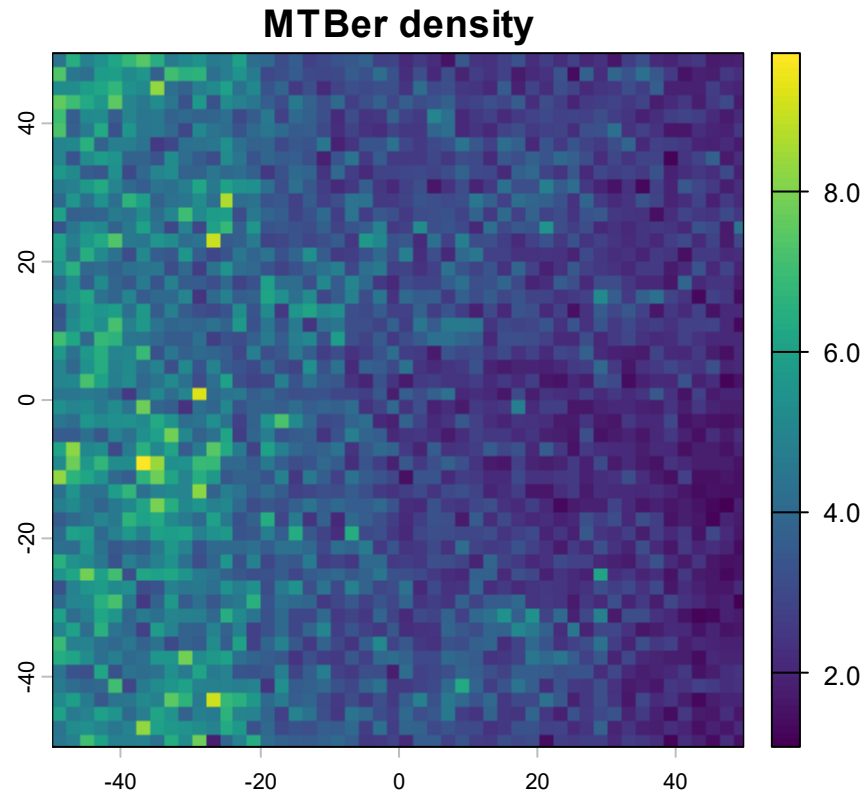
Highest in rural, lowest in towns and in backcountry

We'll simulate car density



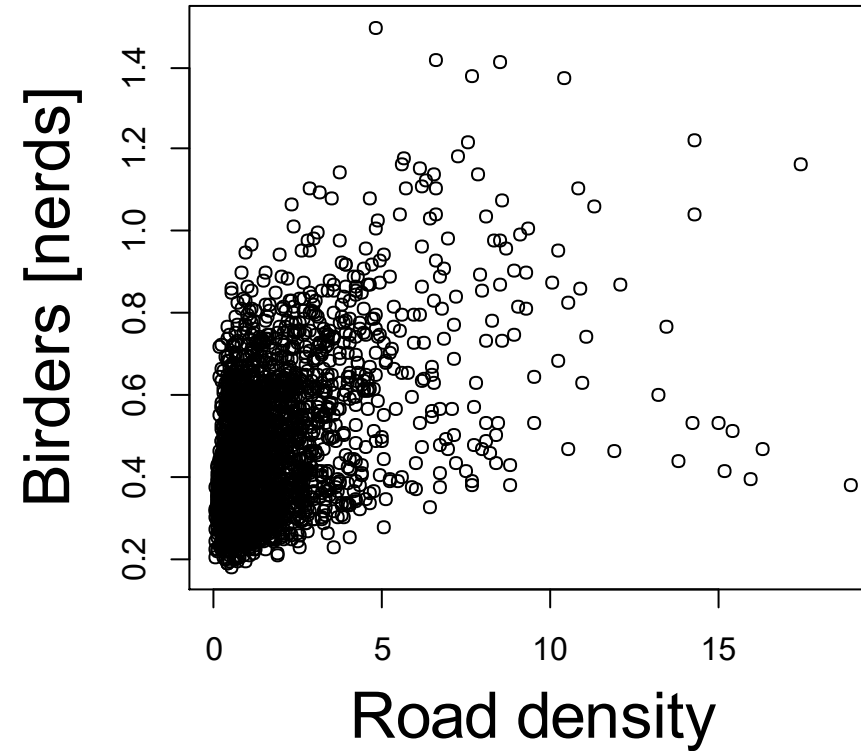
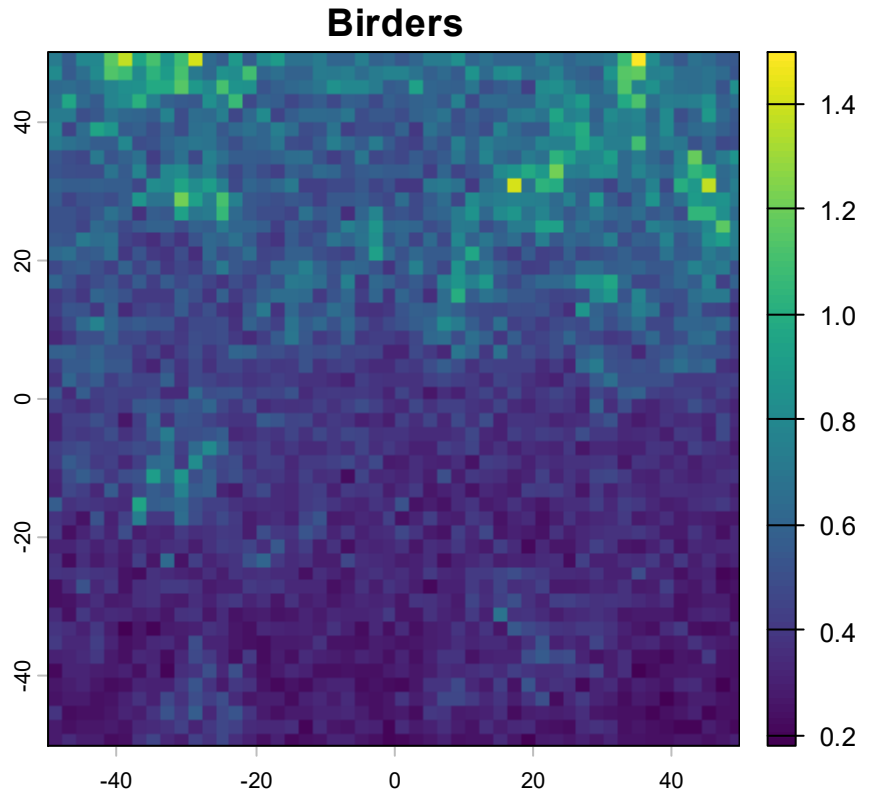
There are cars on the roads...

We'll simulate MTBer density



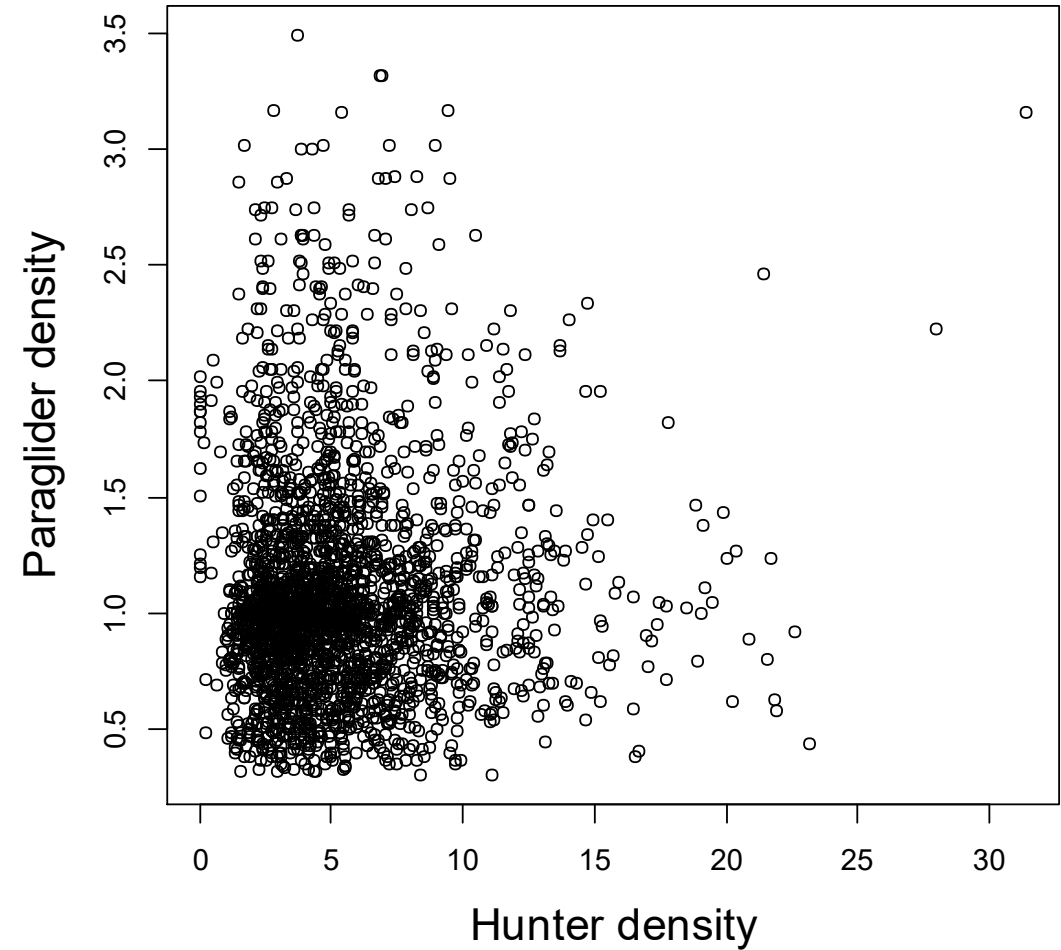
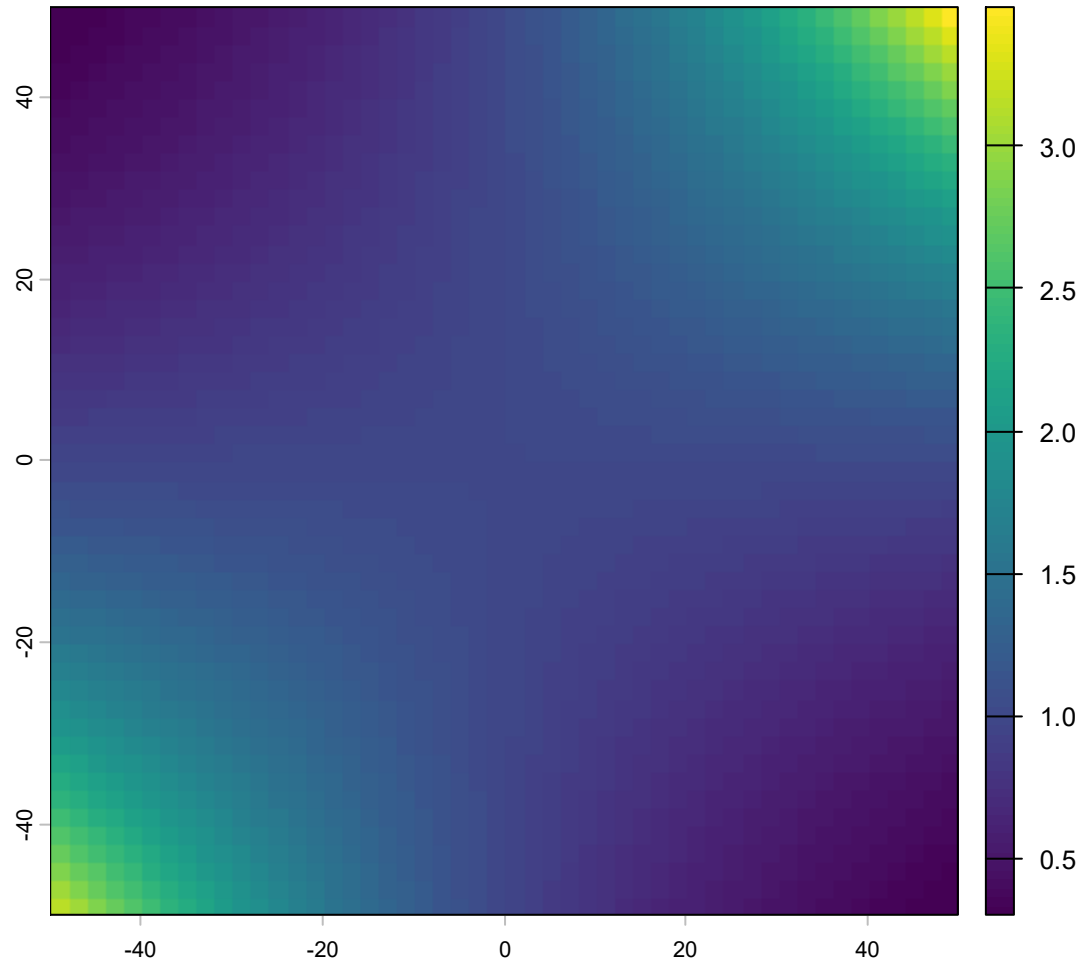
Edge of town, and higher in the west, where the sick trails are

We'll simulate birder density



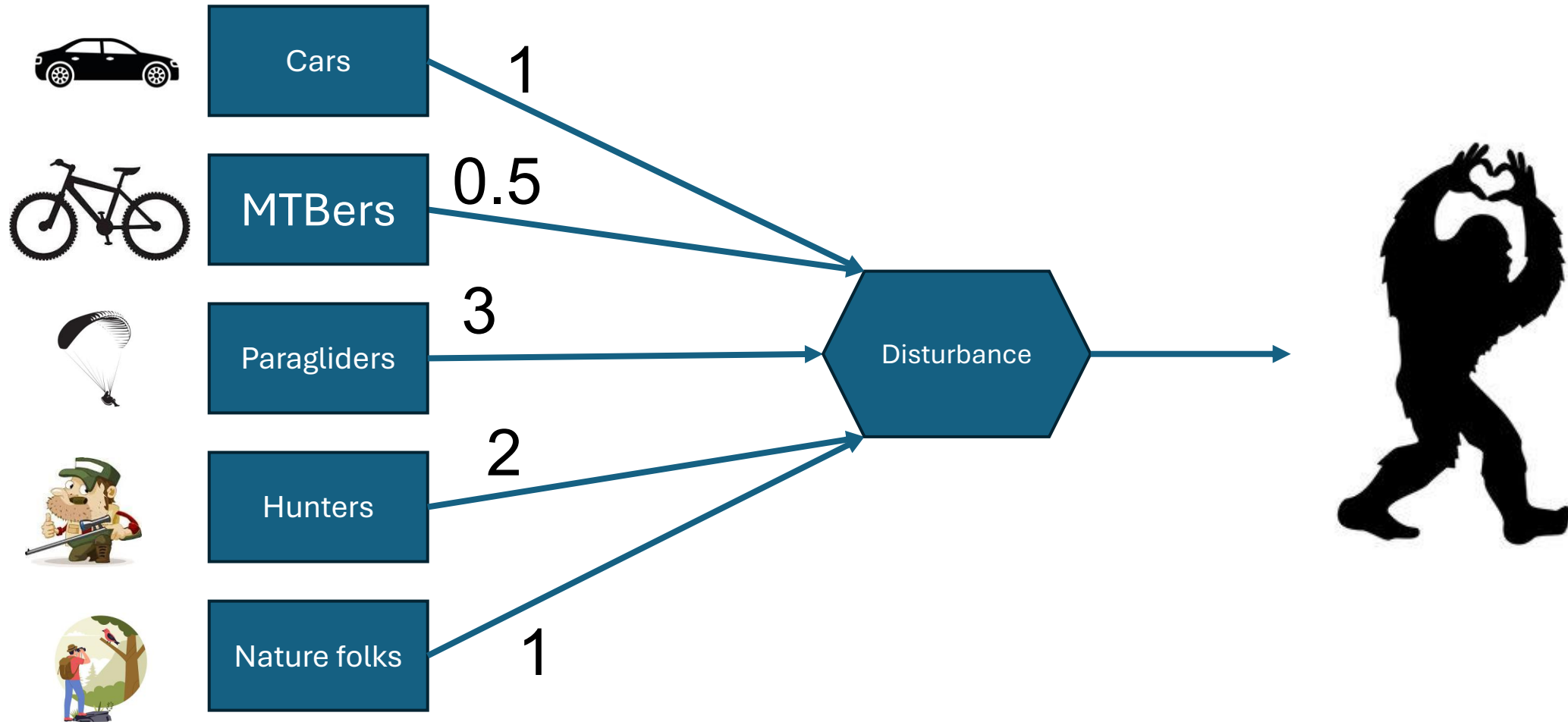
Around town, and in the north, cause it's bird-ier?

We'll simulate paraglider density



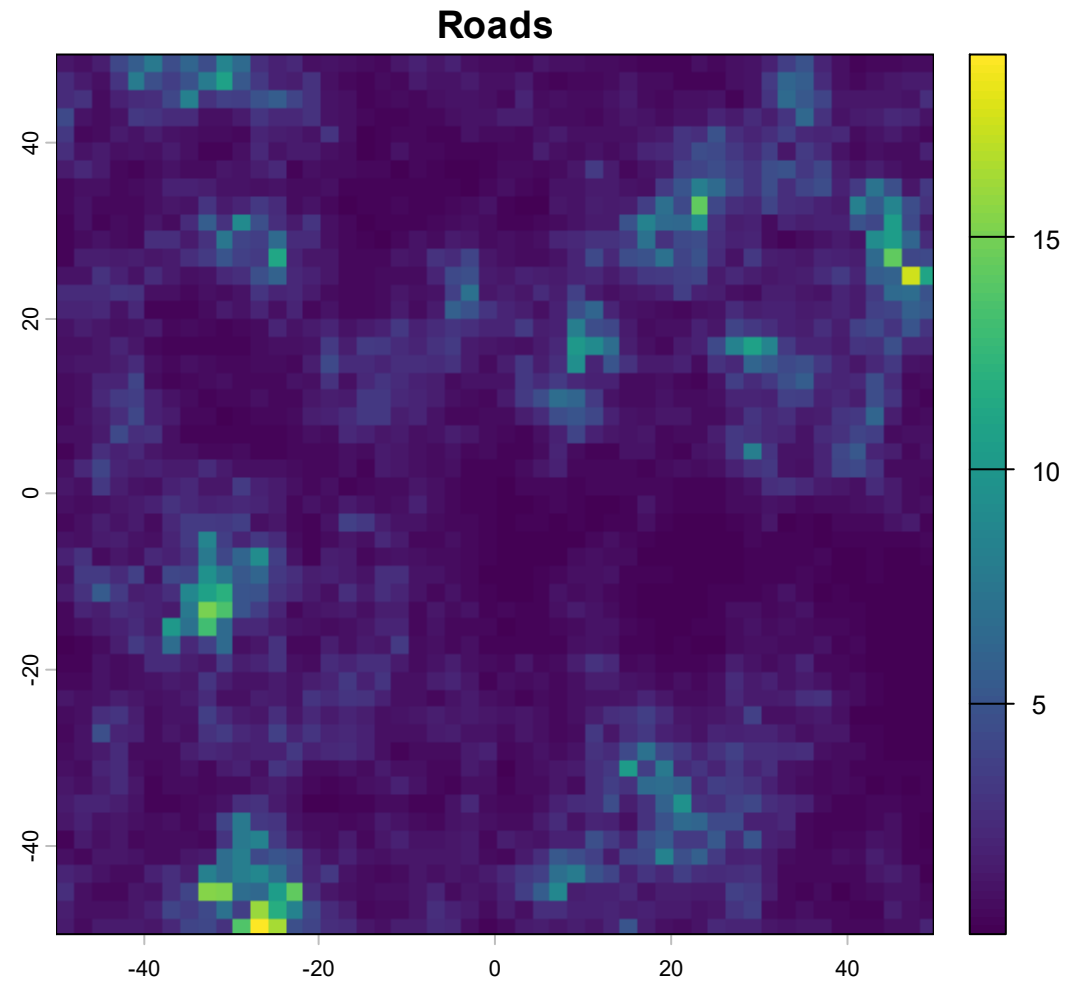
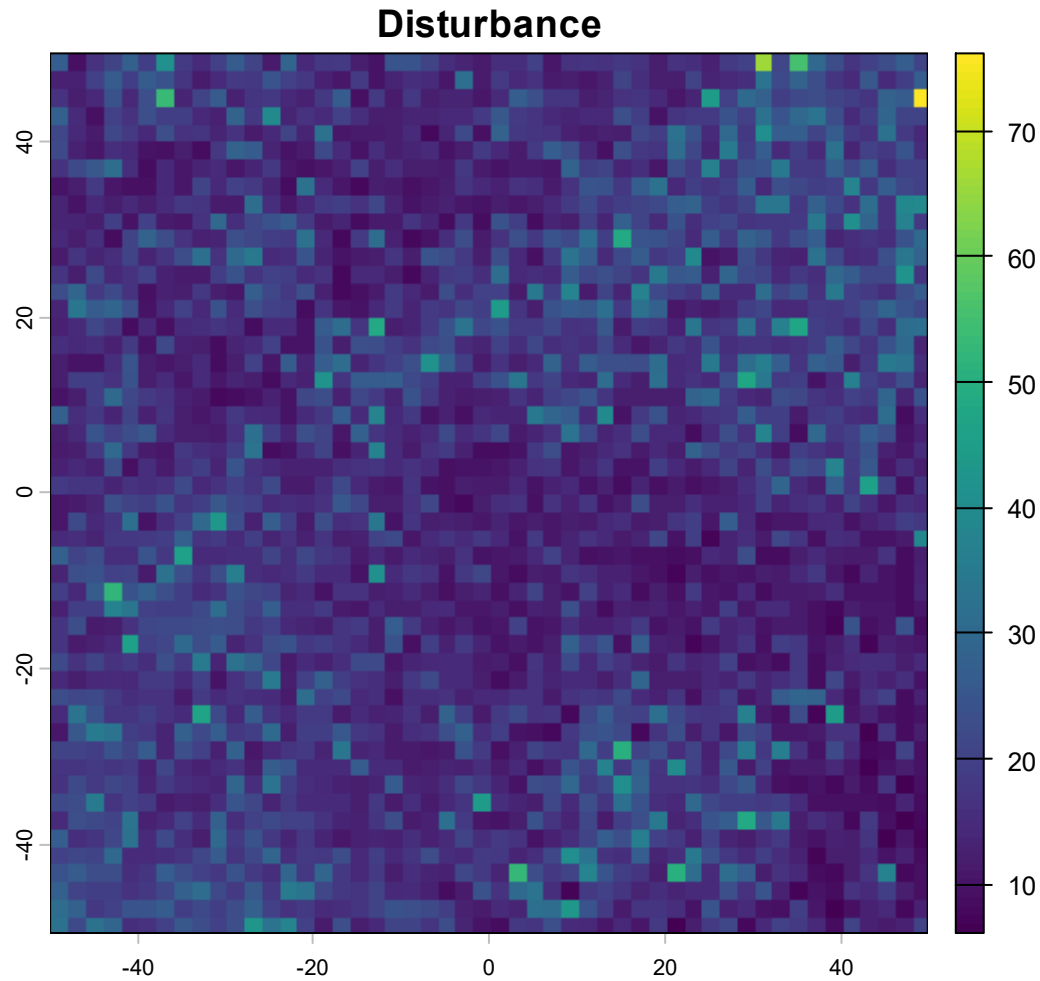
They just launch off two cliffs in the corners

Now we have to make 'disturbance'

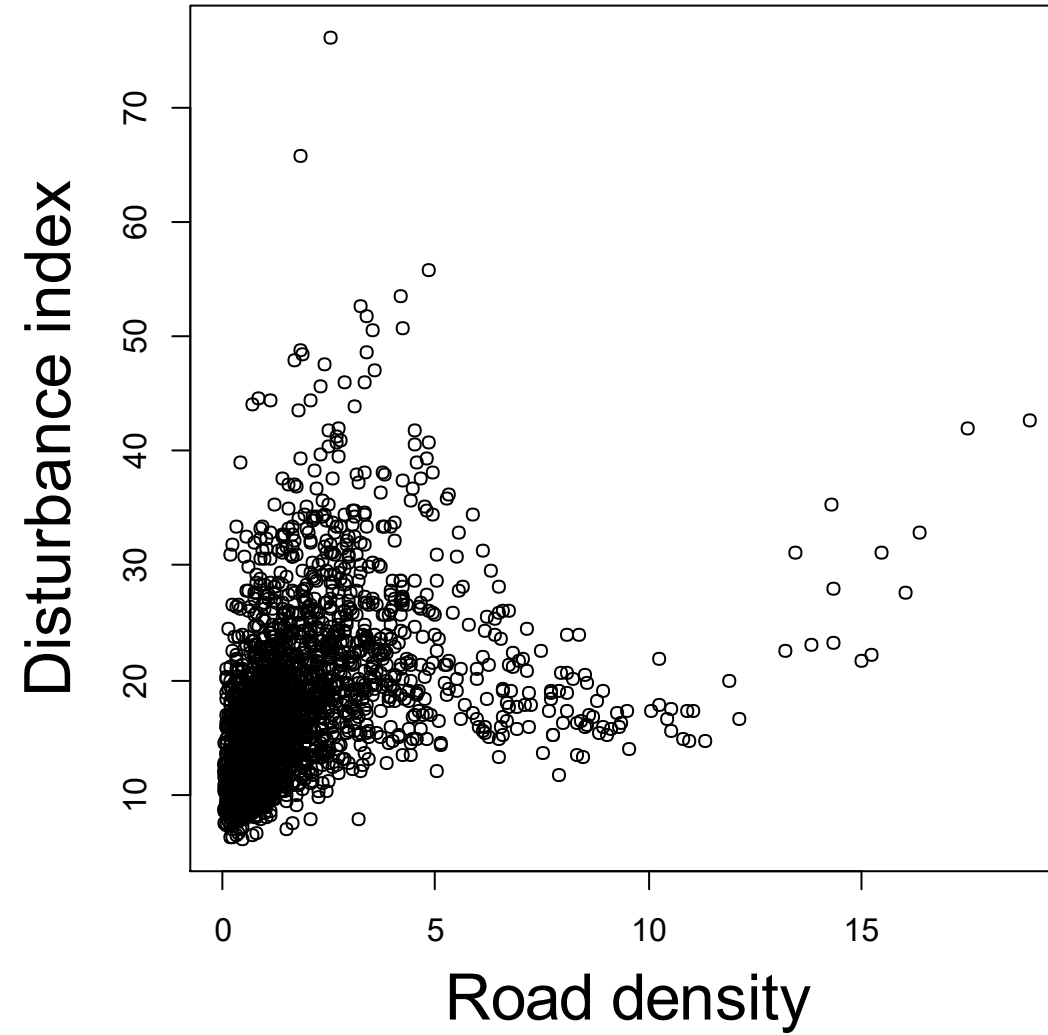


We can incorporate error in our composite covariate...

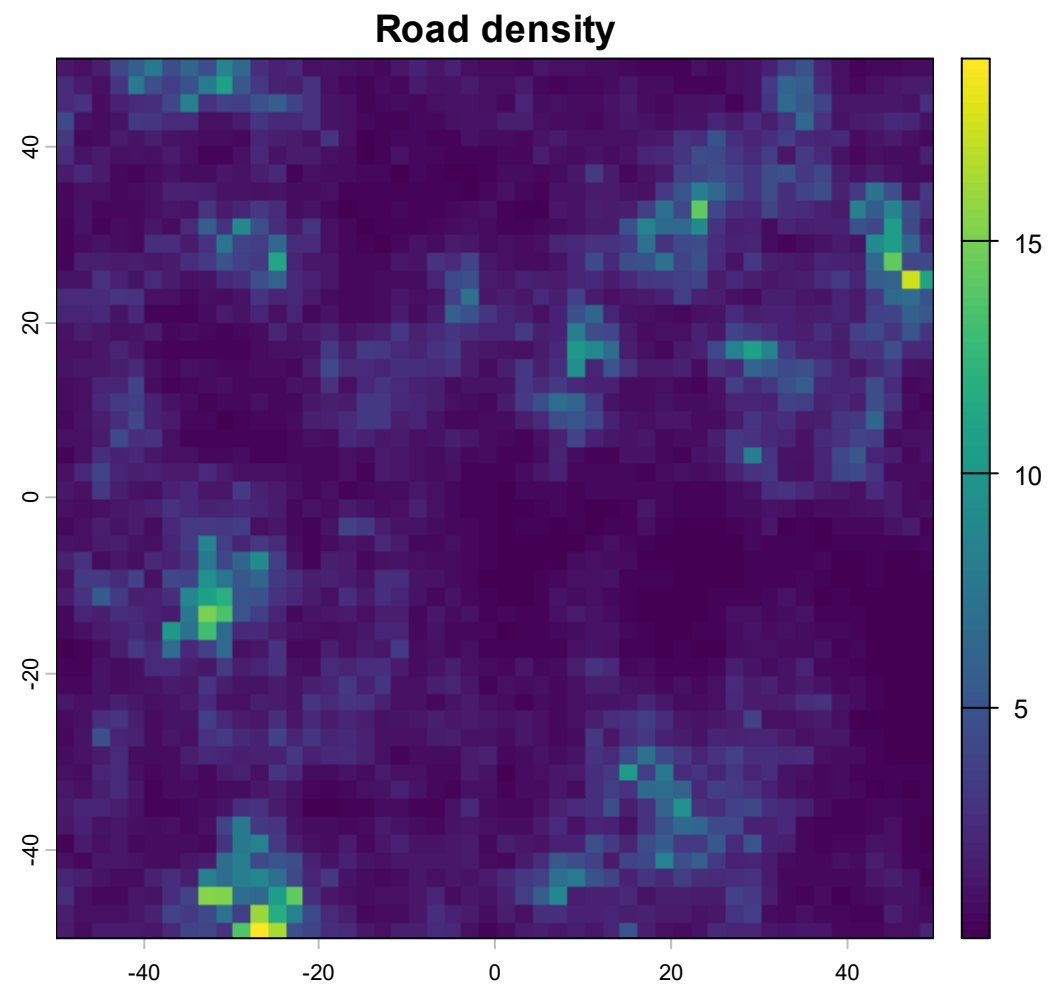
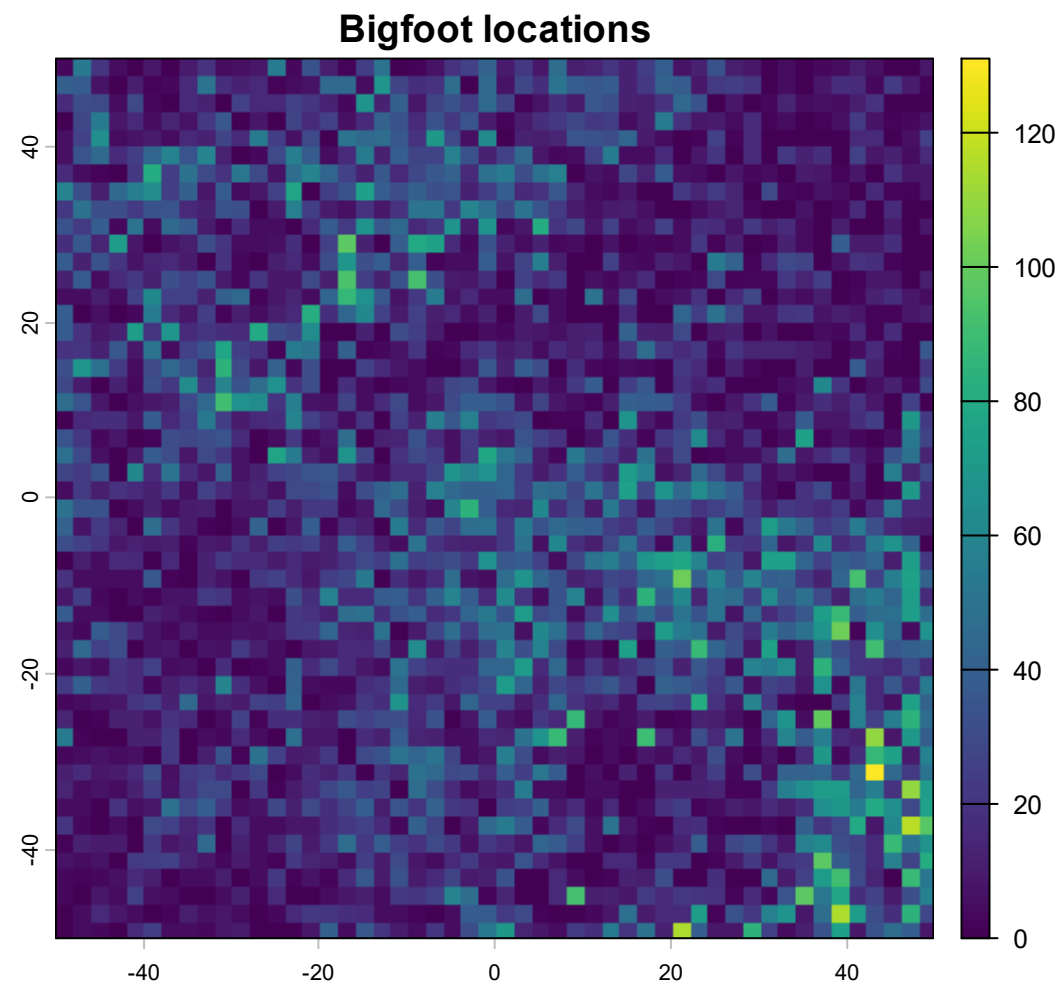
'Disturbance' and road density



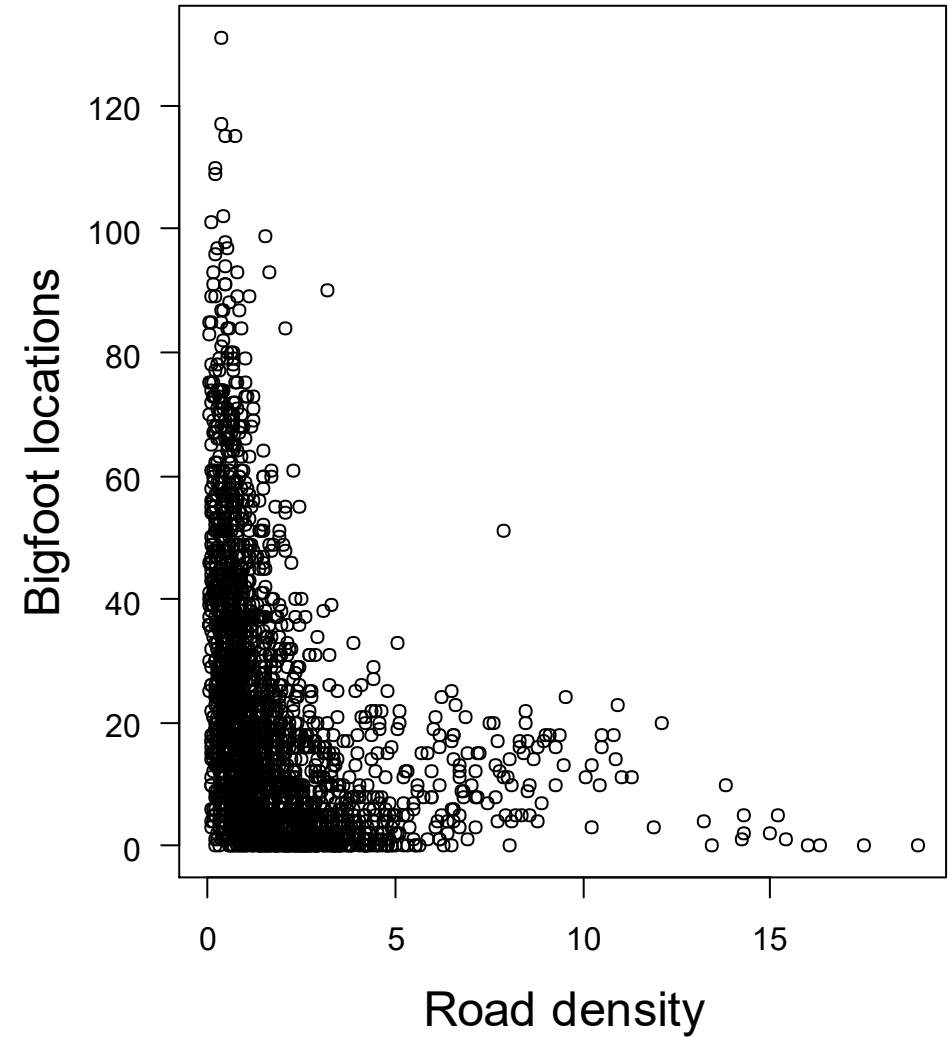
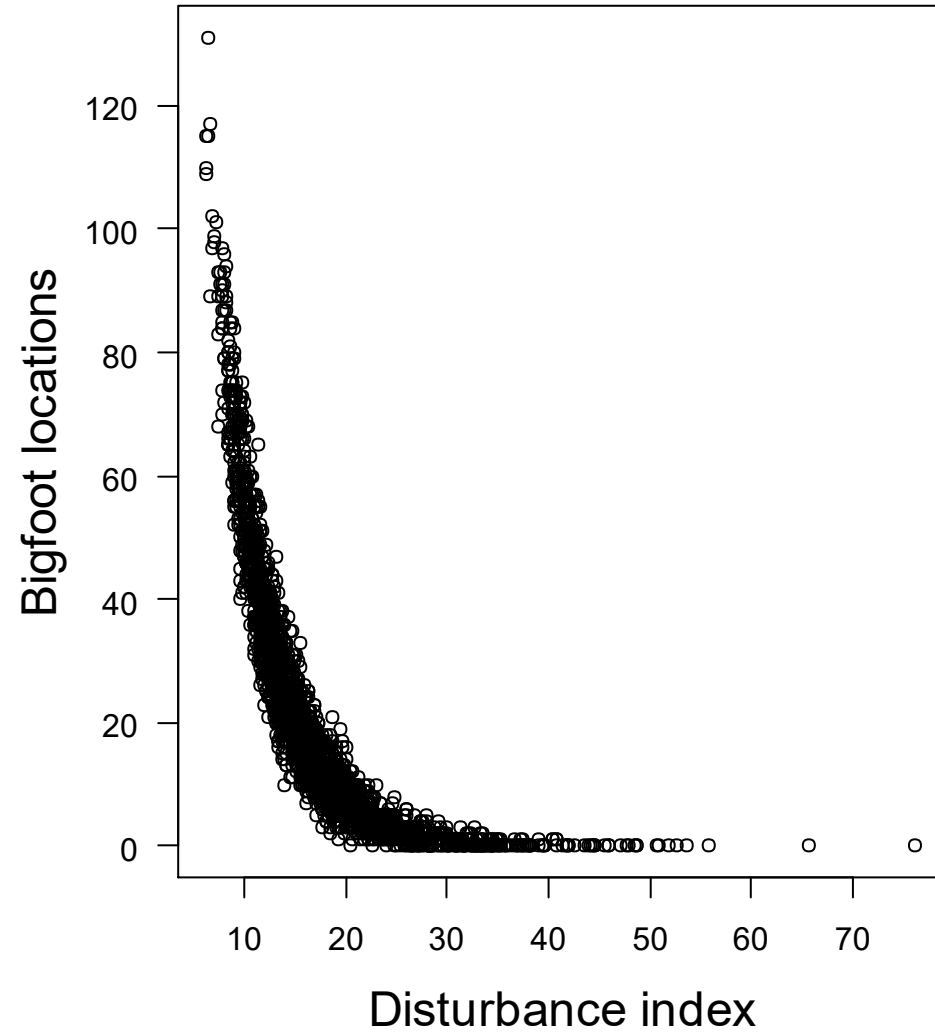
'Disturbance' and road density



Bigfoot use locations



Bigfoot use locations



Building a model

$$\alpha_0 \sim \text{Normal}(5, \sigma^2 = 10)$$

$$\boldsymbol{\beta} \sim \text{Normal}(0, \sigma^2 = 10)$$

$$\alpha_1 \sim \text{Normal}(0, \sigma^2 = 10)$$



$$d_i = 1 \times c_i + \beta_2 \times h_i + \beta_3 \times p_i + \beta_4 \times b_i + \beta_5 \times m_i$$

$$\psi_i = \alpha_0 + \alpha_1 \times d_i$$

$$y_i \sim \text{Poisson}(\psi_i)$$

A quick note...

$$\alpha_0 \sim \text{Normal}(5, \sigma^2 = 10)$$

$$\alpha_1 \sim \text{Normal}(0, \sigma^2 = 10)$$

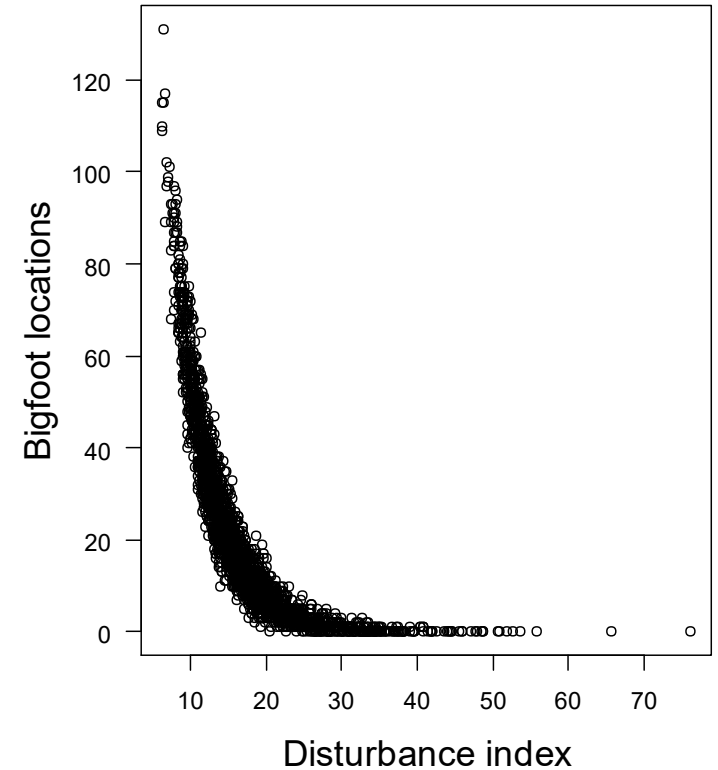
$$\beta \sim \text{Normal}(0, \sigma^2 = 10)$$



$$d_i = 1 \times c_i + \beta_2 \times h_i + \beta_3 \times p_i + \beta_4 \times b_i + \beta_5 \times m_i$$

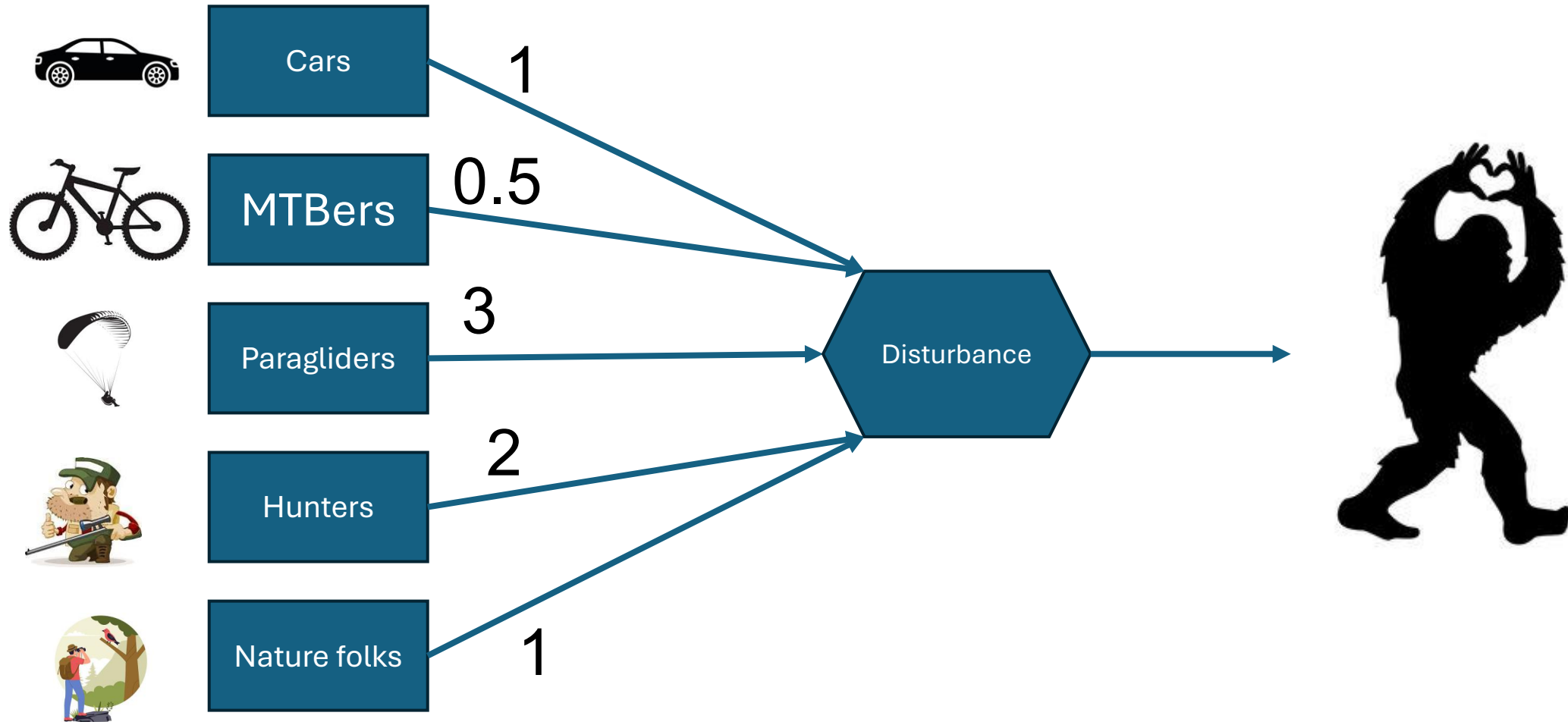
$$\psi_i = \alpha_0 + \alpha_1 \times d_i$$

$$y_i \sim \text{Poisson}(\psi_i)$$



It often helps to z-standardize covariates before composite construction

Composite variables are just that... composites!



You already use them all the time...

